

CENTERIS – International Conference on ENTERprise Information Systems / ProjMAN – International Conference on Project MANagement / HCist – International Conference on Health and Social Care Information Systems and Technologies 2025

AI-powered Chatbot influence in Retail Supermarket Customer Services Delivery.

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Abstract

Artificial Intelligence (AI) technologies have been deemed as increasingly transforming global industries, with especially a growing relevance in addressing service quality challenges in retail sectors within developing economies. This study investigates the influence of AI-powered chatbots on customer service delivery in South African retail supermarkets. Adopting an interpretivist paradigm, qualitative data were collected from online shoppers and retail employees across three major food retailers. Thematic analysis using NVivo revealed that chatbots enhance customer-centric service delivery and enable retailers to extract actionable insights from consumer data. These findings highlight the potential of AI chatbots to improve service quality in resource-constrained contexts. Future research should incorporate quantitative methods, assess long-term effects, and consider ethical frameworks for responsible AI deployment.

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Peer-review under responsibility of the scientific committee of the CENTERIS/ProjMAN/HCist conferences

Keywords: AI-powered chatbots; retail supermarkets; customer services delivery.

1. Introduction

Artificial Intelligence (AI) has profoundly influenced daily human activities through the development of intelligent agents capable of performing diverse functions. Among these, chatbots, AI-powered programs that simulate human-

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like interactions via text or voice [1], have gained prominence. Leveraging Natural Language Processing (NLP) and Human-Computer Interaction (HCI) models [2], chatbots play a pivotal role in service automation. They enhance customer satisfaction by delivering timely, efficient, and conversational support [3]. As AI technology advances, chatbots increasingly replace human agents in customer service roles, streamlining operations, reducing response times, and improving the overall quality of customer interactions [3] [4]. These systems contribute to operational efficiency, cost reduction, and faster service delivery, ensuring prompt and effective customer support [5].

In South Africa, major grocery retailers such as Makro, Woolworths, Checkers, and Pick ‘n Pay operate online platforms to support digital shopping experiences [6]. Despite these advancements, AI-powered chatbot interactions remain limited in their ability to replicate human-like engagement [7]. Research highlights that chatbots often fall short in meeting user expectations for natural, emotionally intelligent communication [8]. While offering operational efficiency, they often lack the emotional intelligence and personal touch that human agents provide [4]. This gap raises important questions about customer satisfaction and trust in AI-based service delivery. The study’s problematisation [9] is grounded in understanding whether customers trust chatbots to resolve their issues effectively and what type of interaction they prefer [10].

The objective of this study was “*To explore the influence of AI-powered chatbots on customer services delivery in the South African retail supermarkets.*” The remaining discussion is structured into five sections: a literature review outlining the historical development of chatbots; a methodology section applying the “research onion” framework to explain philosophical stance, methods, and data collection; a findings section presenting qualitative results; and a discussion addressing ethical, social, and technical challenges. The fifth section details limitations and future research directions, while the final section concludes by summarising key insights and suggesting advancements in chatbot interaction design.

2. Evolution of Chatbot Technology

AI-powered chatbots operate through algorithmic systems that process written user input to generate text and audio responses [11]. Although inherently robotic, recent advancements have enabled chatbots to mimic human-like behavioural traits, particularly in service delivery contexts [8]. The evolution of chatbot technology reflects major progress in artificial intelligence and natural language processing. Early models, such as ELIZA (1966), relied on basic pattern-matching and scripted responses, offering limited conversational depth [12]. Subsequent rule-based systems in the 1990s and early 2000s improved functionality but remained constrained in contextual understanding [13]. The advent of machine learning and deep learning marked a significant shift, allowing chatbots to learn from large datasets and produce more coherent, human-like interactions [14]. Recent innovations, including generative models like OpenAI’s GPT series, have further enhanced chatbots’ ability to understand context, maintain dialogue, and perform tasks with greater human resemblance [15].

2.1 Changes and use cases brought by use of AI chatbots in service industries

AI-powered chatbots have introduced transformative changes across service industries by combining intelligence, urgency, and human-like interaction capabilities [10]. These traits have positioned chatbots as competitive tools in enhancing service delivery. Beyond answering queries, sales chatbots can recommend products based on extensive historical sales data, demonstrating the integration of machine learning and sentiment analysis to foster emotionally engaging interactions [16]. Consider the following use cases for further clarity.

2.1.1 Education Environments

In educational contexts, chatbots enhance e-learning by delivering instructional content and offering personalised assistance. These systems can recall and reiterate previous lessons, support student inquiries, and accumulate data throughout a course to improve pedagogical strategies [17]. Their integration into learning environments facilitates continuous support and engagement, contributing to improved academic outcomes [18].

2.1.2 Health Care

Healthcare chatbots provide personalised medical information, recommend products and services, and offer preliminary diagnoses based on user-reported symptoms. Examples include Your.MD, Ada Health, Sensely, and Buoy Health [19]. During the COVID-19 pandemic, chatbots like Chasey played a vital role in disseminating public health information [20]. Their benefits include supporting informed decision-making, promoting behavioural therapies, encouraging physical activity, and managing somatic disorders with precision comparable to human practitioners [21].

2.1.3 Robotics

Natural language interfaces developed for chatbots have become essential in robotics, enabling more intuitive human-robot interaction. Early innovations, such as the autonomous robot KAMRO, demonstrated the potential of integrating conversational AI into physical systems [22]. Continued advancements have led to more sophisticated applications, enhancing the functionality and accessibility of robotic systems.

2.1.4 General Service Provisions

In e-commerce and retail, particularly in supermarket environments, chatbots have revolutionised customer service by addressing issues such as delayed human response times and irrelevant answers [23]. Chatbots offer 24/7 support, allowing customers to engage outside standard business hours. Systems built using AIML and Latent Semantic Analysis (LSA) utilise FAQ datasets to deliver prompt and accurate responses, thereby improving customer satisfaction and operational efficiency [24].

2.2 Assessing the changes in retail services with the advent of AI chatbots

In the food and retail industry generally, AI-powered chatbots have become integral to streamlining service delivery. Their functionalities include accepting and tracking orders, coordinating delivery logistics, managing reservations, collecting customer feedback [3], and responding to inquiries using crowdsourced FAQs [25].

In the South African retail context, customer engagement and the challenges of chatbot implementation are particularly significant. Chatbots support personalised recommendations, promotions, and loyalty programs, contributing to improved customer orientation [26, p. 7]. However, online shopping platforms tend to serve more affluent communities, leaving many South Africans reliant on physical stores and long queues for essential goods. To mitigate this disparity, retailers have introduced on-demand mobile applications promising same-day delivery. These apps represent an extension of existing online shopping systems offered by major grocery retailers such as Checkers, Woolworths, and Pick n' Pay, and reflect a growing effort to democratise access to digital retail services [27].

2.3 Use of AI chatbots and the context of retail supermarkets in South Africa

Researchers have focused on improving the understanding of chatbots and customer loyalty and user experience in the retail industry. The provision of a favourable customer experience is essential in guaranteeing heightened levels of customer contentment. Consequently, a robust correlation exists between customer satisfaction and the general user experience [28].

2.3.1 Chatbots as Conversational User Interfaces

Chatbots function as conversational user interfaces (CUIs), designed to assist users in achieving specific outcomes through interactive dialogue. Their purpose is often defined by the user or service provider to ensure mutual clarity of expectations. In this study, chatbot technology as examined from the perspective of customer satisfaction promises improving efficiency, thereby reducing uncertainty and customer anxiety levels [29].

2.3.2 User Experience

The user experience with chatbots is examined through social theories such as computer-mediated social interaction and social response theory, which suggest that users apply human social norms to chatbot interactions [30]. Practical dimensions, including conversational intelligence, context retention, interactivity, and adaptability, are also considered [11]. Integrating both social and functional insights into chatbot design enhances engagement, trust, and usability, addressing key challenges in human-computer interaction [8].

2.4 Literature synthesis

AI-powered chatbots enhance customer engagement and support hyper-personalised experiences. However, integration in South Africa faces challenges due to diverse customer needs and technological constraints [26, p. 119]. User experience depends on chatbots' ability to mimic human interaction, manage complex queries, and maintain conversational flow. Although effective for routine inquiries, chatbots struggle with interpreting emotional cues because they lack facial expressions and tone, limiting empathy recognition [31].

Research highlights the potential of chatbots to enhance user experience and customer loyalty when designed with social norms and practical conversational elements in mind, such as maintaining flow and adaptability [7], [10]. By aligning chatbot design with user motivations and expectations, retailers can foster positive experiences that strengthen brand loyalty [28]. Trust in chatbots is influenced by factors such as their behaviour, visual presentation, and user privacy concerns, along with human-like features, including names, personalities, and language processing abilities [32]. As such, the emotional intelligence of chatbots is pivotal for creating human-like interactions, with methods aimed at cultivating empathy within these bots.

3. Research Methodology

This section outlined the research design undertaken to execute this study. Using the Research Onion guide by Saunders and others [33], it followed the theoretical and practical steps in chronology of how the researchers presented them for seamless flow.

3.1. Theoretical underpinnings

3.1.1 Research Philosophy

This study adopted an *interpretivist research philosophy*, which is explanatory in nature [34] and well-suited for exploring how individuals construct meaning within their social contexts. This philosophical stance aligns with the study's objective, to explore the influence of AI-powered chatbots on customer service delivery in South African retail supermarkets by enabling the capture of rich, narrative insights that quantitative approaches might overlook.

3.1.2 Research Strategy

A *case study strategy* was employed to facilitate in-depth exploration of individual experiences through qualitative interviews [35]. This approach allowed for the construction of detailed explanations regarding observed behaviours and patterns within specific retail environments. Each supermarket was treated as a distinct case, and each participant was considered a case in their own right [36], enabling a granular understanding of chatbot interactions and customer service dynamics.

3.1.3 Choices

The study utilised a *multi-method qualitative approach*, integrating various qualitative techniques such as face-to-face interviews, observations, note-taking, and phone calls [37]. This approach enriched the data by capturing diverse perspectives and experiences, thereby enhancing the robustness and depth of the findings [38].

3.1.4 Time Horizon

A *cross-sectional time horizon* was adopted to examine the immediate relationships between chatbot features and customer service outcomes [39]. This approach enabled the study to assess usability and user perceptions at a specific point in time.

3.2. Practical Techniques and Procedures

3.2.1 Target Respondents & Sampling

Data were collected from three selected supermarket contexts in South Africa using interviews, observations, documents, and records. Participants were purposively sampled based on their background, familiarity with AI-powered chatbots, and direct experience with both chatbot and in-person retail interactions. This purposive sampling strategy ensured the relevance and richness of insights gathered from each case [38].

3.2.2 Data collection and analysis

Semi-structured interviews and observations were employed to explore customer interactions with chatbots, focusing on perceived usefulness, ease of use, and satisfaction. While interviews provided reconstructed narratives, observations offered contextual depth. A reflexive thematic analysis (RTA) approach was used to identify recurring patterns and develop initial and final themes. Meaning-based interpretation was guided by five dimensions, *sense, feel, think, act, and relate*, adapted from prior research [40] [41]. Interview data were transcribed and analysed using NVivo, combining open coding and researcher-driven sense-making within the RTA framework [42].

3.2.3 Ethical Considerations

Reflexivity and researcher bias were acknowledged as inherent to qualitative research [43], with subjectivity considered a valuable analytical lens [44]. In other words, the analysis could not be separate from the “knowing” researcher [40]. Ethical clearance was obtained from the relevant Research Ethics Committee prior to data collection. Informed consent was secured from all participants, and anonymity was maintained.

4. Presentation of findings and analysis

The study utilised purposive sampling, a widely recognised method in qualitative research, to identify and incrementally select key participants on whose experiences, findings were extracted. The decision to stop interviewing was taken at participant 10, as the study had gathered adequate qualitative data to conduct a conclusive analysis of the insights.

4.1. Descriptive results

In terms of demographic representations, the number of females to males was equal (5 apiece), ages ranging from 18 to 56 years of age, 9 nationals of South Africa and 1 non-South African National. These participants held qualifications of high school graduate to master’s degree between themselves. All these qualities were necessitated by the assumption that if respondents are heterogenous in nature, the findings will present a rich outcome.

Table 1. Demographical Data of Participants.

ID	Gender	Age range	Education	Job Status	Shopping freq. (w/AI)	Nationality
P-1	Female	18-24	High School	Student	1-2 times a month	South African
P-2	Female	41-56	Master's	Employed	1-2 times a week	South African
P-3	Female	25-40	Bachelor's	Employed	1-2 times a month	South African
P-4	Male	41-56	Master's	Employed	1-2 times a week	South African
P-5	Female	41-56	Bachelor's	Employed	3+ times a week	Non-South African
P-6	Male	25-40	Bachelor's	Employed	1-2 times a month	South African
P-7	Male	25-40	Bachelor's	Employed	3+ times a week	South African
P-8	Female	18-24	Bachelor's	Employed	1-2 times a month	South African
P-9	Male	18-24	Bachelor's	Employed	3+ times a week	South African
P-10	Male	41-56	Bachelor's	Employed	1-2 times a month	South African

Table 1 shows the online shopping frequency of participants. It shows that the majority of participants shop online 1 to 2 times a month, with fewer participants shopping at least 3 times a week, and even fewer shopping 1 to 2 times a week. This distribution indicates a varied but primarily moderate level of online shopping activity among the group.

4.2. Explanatory Results

The study employed Reflexive Thematic Analysis (RTA) to interpret qualitative data, following Braun and Clarke's six-phase framework: familiarisation, coding, theme development, review, definition, and reporting. An inductive, data-driven approach was used to generate initial codes from anonymised transcripts (P1–P10). Themes were constructed around central organising concepts and refined through iterative analysis. Final themes were defined and named based on their significance, forming the basis for the study's interpretive narrative. NVivo software supported open coding and meaning-based interpretation [40], [42] across five dimensions of *sense, feel, think, act, and relate*.

4.3. Formulation of Themes

Using investigator discretion [44] and based on the initial coding and analysis of the data, 9 sub themes were formulated. The sub themes are customer satisfaction, customer needs, AI chatbot services, AI chatbot sociability, AI chatbot challenges, improvement in SA retail market, AI chatbot trustworthiness, post-live bot chat feedback, and online shopping influences. These themes are depicted in Figure 1.



Fig. 1. Formulated themes.

Through Reflexive Thematic Analysis, three overarching meaning-based themes were developed to directly address the research question: “How do AI-powered chatbots influence customer service delivery in South African retail supermarkets?” These themes are:

1. *Customer Service Delivery* – derived from sub-themes of customer satisfaction, customer needs, and chatbot trustworthiness.
2. *AI-Powered Chatbot Services* – informed by chatbot sociability, service capabilities, and associated challenges.
3. *Changes to Retail Supermarkets* – shaped by post-chat feedback, the influence of online shopping, and improvements in the South African retail landscape.

These themes collectively offer a structured understanding of how chatbot technologies are reshaping customer service experiences in the retail sector.

4.3.1 Customer services delivery

The respondents inferred in unison that there needs to be customer satisfaction in the way chatbots engage with clients who turn to them for clarity about products when human agents are not available. This was said to be what is the prime need for customers, a development that was deemed will increase trustworthiness in the customer-chatbot relationship.

“OK, so I'd give it a 5 out of five because. Of the easy access to products because of new notifications that pop up.” P-1

“Yeah, I think the general needs are met.” P-3

“Yes. Yeah, it is doing a great job. Yes, I do. I do feel like it's a good idea 24/7...” P-6

“I trust it fully. I think it's, you know, I think also being a technologist or someone within it, understanding how these systems are built. I have full trust in it.” P-7

4.3.2 AI-powered chatbot services

The ability for AI-powered chatbots to emulate humanlike qualities made customers feel they can be sociable with them. Even though they were some challenges, customers were hopeful that they will be sociable one day. AI-powered chatbots were always ready to serve round the clock. To be AI-powered meant they can be available at any time of the day.

“I think there will be two things, one is the one that I've already mentioned to say, if they can make the app user friendly for, you know, most people, obviously it can all the people who use it.” P-2

“Definitely what I'd like to see going forward is that sociable conversation where I tell the system, hey, I'm looking for an invoice for my purchases in May. Oh, OK, here it is. Off you go.” P-4

“Chatbots are used primarily for FAQs and immediate assistance, particularly in online shopping scenarios.” P-8

“Yeah, I [I] could see it. It does care about my needs ... Users often rely on chatbots for customer service issues, preferring them for quick responses rather than traditional email support.” P-9

4.3.3 Changes to retail supermarkets

There is an open secret that the chatbot technology has become a game changer in the South African retail supermarket space. Users typically engage with chatbots to inquire about products they are unfamiliar with or to gather information about items and services, something they would not do with human agents on a regular busy day.

“For example, some users look for guidance on new products.” P-1

“...or details about specific items.” P-3.

“Unlike pushing [a] trolley going through aisle to aisle, sometimes not even finding what you're looking for. So, I feel like now I prefer shopping more online than in the physical store.” P-9

“Pre COVID I would really go to the store directly, you know the physical store and buy stuff, but after COVID there was really no need for me to go to the store because you know, the [AI] technology ... helps me with free delivery ... the

products, and interacting with this door through the technology has helped me not to interact with this door physically.”
P-10

5. Discussion

This section presents a thematic discussion based on the Reflexive Thematic Analysis of ten in-depth interview transcripts. Despite the relatively small sample size, the case study approach enabled rich, detailed narratives, with data saturation observed by the midpoint of data collection. The analysis explores: (i) the influence of AI chatbots on shopper behaviour, (ii) enhancements to customer shopping experiences, and (iii) the challenges and opportunities of chatbot adoption in South African retail. The section concludes with a reflection on study limitations and directions for future research.

5.1. AI-powered chatbot influence on shoppers

Although participants did not always explicitly recognise the role of AI-powered chatbots, qualitative analysis revealed their substantial influence on customer service. Interview data indicated that chatbots enhance efficiency, convenience, and user experience by facilitating product access, delivering accurate information, and providing personalised recommendations. Participants appreciated comprehensive support and recommended broader service coverage and customer education to boost adoption. Despite some indifference or scepticism, findings suggest chatbots positively affect consumer behaviour, warranting further research using larger and longitudinal datasets to determine the extent of this impact.

5.2. Enhancing customer experience through AI-powered chatbots

AI-powered chatbots significantly enhance customer experiences by delivering timely, human-like responses. Literature highlights that attributes such as personalised naming, conversational flow, and emotional responsiveness improve engagement and satisfaction [4], [16], [23], [25], [29], [31], [45]. Through natural language processing and sentiment analysis, chatbots can adapt to customer emotions, influencing purchasing decisions and fostering loyalty [3]. Their ability to operate continuously, manage multiple queries, and support multilingual interactions contributes to reduced operational costs and improved accessibility [46]. Psycholinguistic insights suggest future chatbot development should focus on replicating human speech and thought processes to deepen interaction quality [47]. Key implications include the economic potential of chatbots in South African supermarkets, their role in building trust [3] and empathy [48], and the need to address technological limitations [15] to ensure broader adoption and effectiveness.

5.3. Application in South African Retail: Opportunities and Challenges

AI-powered chatbots are increasingly contributing to efficient and accessible customer service in South Africa's major supermarket chains [26]. Their adoption offers benefits such as real-time support, streamlined order processing, and enhanced engagement through personalised services [4], [45]. However, local challenges, including linguistic diversity, cultural nuances, and uneven technological infrastructure, necessitate context-sensitive adaptations [26], [28]. To ensure broader acceptance and effectiveness, chatbots must incorporate multilingual capabilities and culturally relevant responses tailored to South Africa's diverse demographic landscape.

5.4. Limitations and Future Research Directions

Despite the potential of AI-powered chatbots to enhance customer service, limitations persist in replicating human-like empathy and handling complex, emotionally charged interactions [7], [49], [50]. Chatbots perform well with routine queries but struggle with trust-building and emotional engagement, particularly among less technologically savvy or lower-income demographics [51]. Future research should focus on advancing conversational AI, improving

personalised recommendations, and integrating human support for complex queries. In the South African retail context, addressing these limitations is essential for broader adoption and effective implementation.

6. Conclusion

The study examined chatbot-user interactions, identifying strengths and areas for improvement, particularly in personalisation and ethical design. Users generally perceived chatbots positively when they were intuitive, user-friendly, and humanlike. Qualitative findings highlighted perceived usefulness, defined by urgency, accuracy, availability, access, and ease of use, as a key factor influencing satisfaction. While users valued efficient query resolution and assistance, concerns persisted regarding nuanced understanding. Semi-structured interviews revealed that conversational flow, response accuracy, and adaptability significantly shape engagement. Humanlike responses and memory of prior interactions were especially favoured for contextual relevance.

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